

TECH NEWS

TECHNOLOGY UPDATES

Ultrasound to help deliver drugs

Researchers are fitting people's heads with ultrasound-emitting helmets to treat tremors and Alzheimer's. They are using it to remotely activate cancer-fighting immune cells. Startups are designing swallowable capsules and ultrasonically vibrating enemas to shoot drugs into the bloodstream. One company is even using the shockwaves to heal wounds. All this with the help of modern-day medical imaging and bubbles.

Bubbles are nothing but gas-filled microbubbles that technicians use to bump up contrast in grainy ultrasound images.

"We're starting to realise they can be much more versatile," says Tao Sun, a 27-year-old electrical engineering student who studies microbubbles. "We can chemically design their shells to alter their physical properties, load them with tissue-seeking markers, even attach drugs to them."

Nearly two decades ago, scientists discovered that microbubbles could shake loose the blood-brain barrier—the reason why 98 per cent of drugs simply can't get to the brain. If you station a battalion of microbubbles at the barrier and hit them

with a focused beam of ultrasound, the blood-brain barrier opens—and for a few hours, any drugs that happen to be in the bloodstream can also slip in. However, with too much pressure, those bubbles can implode violently, irreversibly damaging the barrier.

That's where Sun has developed a device that could listen in on the bubbles and tell how stable they were. If he eavesdropped while playing with the ultrasound input, he could find a sweet spot where the barrier opens and the bubbles don't burst. Current technology requires patients to wear special ultrasound helmets and hop in an MRI machine, to ensure the sonic beams go to the right place.



At the Sunnybrook Institute in Toronto doctors have begun using focused ultrasound, delivered through helmet-like devices such as this, to loosen the blood-brain barrier, making it possible to deliver life-saving drugs

Hyperloop coming to life in Nevada

Elon Musk's idea of the hyperloop is taking shape in the Nevada desert, where Virgin Hyperloop One is building a giant version of the futuristic, tube-based transportation system.

The big idea reached the world when Elon Musk published a 57-page white paper outlining his thinking for firing levitating pods, carrying passengers or cargo through nearly airless tubes, at speeds up to 1126kmph (700mph). The brilliance of the hyperloop by Virgin Hyperloop One is that it is just a collection of existing transportation and industrial technologies. It's a chimera, part elevated structure, metal tube, bullet train, pressure vessel and vacuum system, all smooshed together. Hyperloop One expects to launch a commercial hyperloop system in 2021.



Virgin Hyperloop One wants to launch a commercial hyperloop in 2021, and it's busy in the Nevada desert trying to make it happen